Product Note: Card Spend AI

Failure Modes

1. **Unknown merchants** — the biggest gap. Any merchant not in the database falls to keyword or hint inference, producing low-confidence recommendations.
2. **Noisy merchant strings** — credit card statements produce strings like "GRAB* A-4KXYZW". Preprocessing catches common patterns but new formats break silently.
3. **Ambiguous category hints** — "travel" could mean airlines, hotels, or a travel portal, each with different reward rates. We check all sub-categories and pick the best reward, but the user doesn't know we made that choice.
4. **Close card calls** — when two cards are within 20% reward rate, the "best" card depends on factors we don't model: spending caps, sign-up bonuses, monthly minimums.

What to Log

Every recommendation should store: raw merchant string, normalized string, resolved category, match type and confidence, fuzzy score (when applicable), category hint vs. category resolved, all cards ranked with rates, and whether `update_needed` was triggered. This enables two feedback loops: improving merchant coverage and improving recommendation quality.

How to Improve Coverage

Build a merchant-learning loop. Every low-confidence or missing-merchant case gets logged, reviewed, and promoted into the merchant database. This is the highest-leverage investment because:

- The system's main failure mode is merchant coverage
- Better merchant resolution improves every downstream recommendation immediately
- It turns uncertain runtime guesses into permanent knowledge-base improvements
- It's much more practical than building a smarter model from scratch

The workflow: system flags uncertain cases → reviewer approves/corrects → merchant and aliases added to database → future queries hit exact/fuzzy match instead of inference.

2-Week Roadmap

Week 1: Merchant learning pipeline — Build a review UI for the update queue (approve/reject/correct). Add approved merchants and aliases back into the database automatically. Track approval rate to measure inference quality.

Week 2: Recommendation quality — Add user feedback capture (did you use this card?). Replace keyword matching with sentence-transformers for better unknown merchant categorization. Add spending cap awareness to card ranking.

The merchant learning pipeline comes first because it compounds: every merchant added improves all future recommendations with zero model changes.